

CSES Introductory Problems — Problem Reference

All listed tasks: 1 s, 512 MB. Examples omitted for compactness.

1072 — Two Knights

For every $k = 1, 2, \dots, n$, count ways to place two knights on a $k \times k$ chessboard so that they do not attack each other.

In: n

Out: n integers: the answer for each $k = 1, \dots, n$

Constraints: $1 \leq n \leq 10^4$.

1618 — Trailing Zeros

Given n , find the number of trailing zeros in $n!$. **In:** n

Out: number of trailing zeros in $n!$

Constraints: $1 \leq n \leq 10^9$.

1622 — Creating Strings

Given a lowercase string, generate every distinct permutation of its characters. **In:** one lowercase string s

Out: first the number k of distinct strings; then all k strings in lexicographic order

Constraints: $1 \leq |s| \leq 8$.

1623 — Apple Division

There are n apples with weights p_i . Partition all apples into two groups, minimizing the absolute difference between the groups' total weights. **In:** n ; then $p_1 \dots p_n$

Out: minimum possible weight difference

Constraints: $1 \leq n \leq 20$, $1 \leq p_i \leq 10^9$. Use 64-bit.

1624 — Chessboard and Queens

Place eight queens on an 8×8 board so no two attack each other. Queens may be placed only on free squares; reserved squares do *not* block attacks. **In:** 8 rows of 8 characters: $\cdot$ free, $*$ reserved

Out: number of valid placements

Constraints: Fixed 8×8 board and exactly 8 queens.

1625 — Grid Paths

A path on a 7×7 grid starts at the upper-left square and ends at the lower-left square, visiting each square at most once. Its 48 moves are described by D, U, L, R, or ? (any direction). Count paths matching the description. **In:** one 48-character string over ?DULR

Out: number of matching paths

Constraints: Grid is fixed at 7×7 ; description length is 48.

1743 — String Reorder

Rearrange the characters of an uppercase string so no two adjacent characters are equal. Among all valid rearrangements, find the lexicographically smallest. **In:** one string s over A–Z

Out: lexicographically smallest valid rearrangement, or -1 if impossible

Constraints: $1 \leq |s| \leq 10^6$.

2165 — Tower of Hanoi

Move n disks from stack 1 (left) to stack 3 (right), using stack 2. Move only the top disk of a stack; never place a larger disk on a smaller one. Minimize the number of moves. **In:** n

Out: minimum move count k , then k lines $a \ b$ describing each move

Constraints: $1 \leq n \leq 16$.

2205 — Gray Code

Construct a Gray code of length n : all 2^n bit strings of length n , ordered so consecutive strings differ in exactly one bit. **In:** n

Out: 2^n lines forming any valid Gray code

Constraints: $1 \leq n \leq 16$.

2431 — Digit Queries

Consider the infinite string 123456789101112.... For each query, find the digit at 1-indexed position k . **In:** q ; then q values k

Out: one digit per query

Constraints: $1 \leq q \leq 1000$, $1 \leq k \leq 10^{18}$.

3217 — Knight Moves Grid

A knight is on an $n \times n$ chessboard. For every square, determine the minimum number of knight moves needed to reach the top-left square. **In:** n

Out: $n \times n$ matrix of minimum move counts

Constraints: $4 \leq n \leq 1000$.

3311 — Grid Coloring I

An $n \times m$ grid contains A, B, C, or D. Change *every* cell to one of these four characters, different from its original character, so no horizontally or vertically adjacent cells are equal. **In:** $n \ m$; then n grid rows

Out: any valid final grid; if none exists, IMPOSSIBLE

Constraints: $1 \leq n, m \leq 500$.

3399 — Raab Game I

Each of two players has cards $1, 2, \dots, n$ and plays each card exactly once. Each round, the higher card scores one point; equal cards score neither. Given final scores a, b , construct a possible play order for both players. **In:** t ; then t cases $n \ a \ b$

Out: for each case, NO if impossible; otherwise YES and two permutations of $1, \dots, n$ realizing the scores

Constraints: $1 \leq t \leq 1000$, $1 \leq n \leq 100$, $0 \leq a, b \leq n$.

3419 — Mex Grid Construction

Construct an $n \times n$ grid. Each cell must contain the smallest nonnegative integer that appears neither to its left in the same row nor above it in the same column. **In:** n

Out: the required $n \times n$ grid

Constraints: $1 \leq n \leq 100$.
